

Answer Key for Strawberry DNA Lab

Part I: Questions

1. What was the purpose of mashing up the strawberry?

To break down the cell wall, cellular and nuclear membranes.

2. What does the extraction buffer do? (Hint: Extraction buffer contains soap. What does soap do when you wash your hands?)

The extraction buffer helps to release the DNA from the surrounding cell components of the crushed strawberry.

3. What does the filter do?

The filter removes the larger particles from the solution, such as seeds, pith, etc., allowing only the smaller cell components such as the DNA, proteins, etc. to filter through.

Questions Parts II and III:

1. What happened when you added the filtrate to the alcohol?

The DNA precipitated out of the solution.

2. What did the DNA look like?

The DNA looked like white, thin fibers wadded up together forming a clump.

Analysis:

3. What does DNA look like?

The DNA will look like a white, cloudy or fine stringy substance.

4. A person cannot see a single cotton thread four classrooms away. But if you wound thousands of threads together into a rope, it would be visible at the same distance. How is this statement an analogy to our DNA extraction?

DNA is not visible as a single strand to the naked eye, but when thousands of threads of DNA are present, you will be able to see the large groups of threads of DNA.

5. Is DNA found in all living or once living cells?

Yes, DNA is present in all living and once living cells. Scientists can use DNA from mummies or seeds that are thousands of years old and identify the genetics of those plants and animals using their DNA.

6. Since the strawberries were once living, and we extracted DNA from them, what does this mean about the foods you eat?

All the foods we eat that come from plants and animals that contain DNA. DNA has been consumed on a daily basis since the beginning of time and is completely safe to eat.

7. Look at the plant cell pictured on page one of the lab handout. Remember that genes are found on chromosomes, and genes control traits. Give at least two examples of traits that are expressed in the strawberry.

Examples: Color, size and sweetness.